

Unit 13, Lesson 14: Using Coordinate Geometry to Solve Real-World Problems



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.



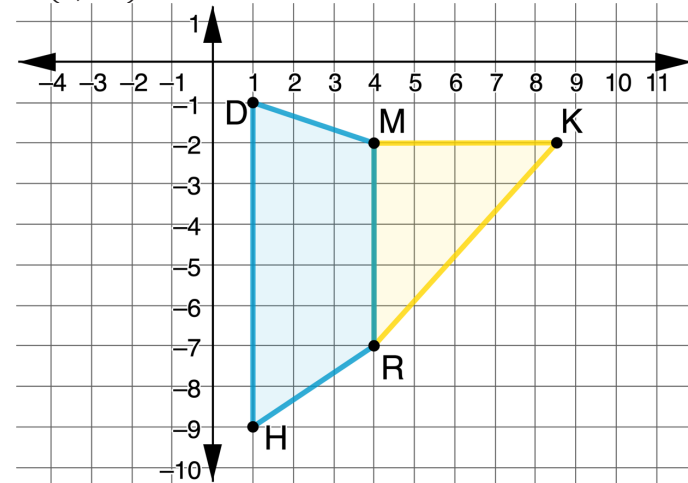
Learning Targets

- ☐ I can use coordinate geometry to classify triangles and quadrilaterals.
- ☐ I can use coordinate geometry to solve real-world problems on the coordinate plane involving perimeter or area of polygons.
- ☐ I can use coordinate geometry to solve real-world geometric problems involving circles, triangles, and quadrilaterals.



13.14.1 Warm-Up: Bell Work

1. The composite figure shown is created by $\triangle MKR$ and trapezoid $DMRH$. $M(4, -2)$, $K(8.5, -2)$, $R(4, -7)$, $H(1, -9)$, and $D(1, -1)$.



- a. Complete the table. If necessary, round to the nearest thousandth.

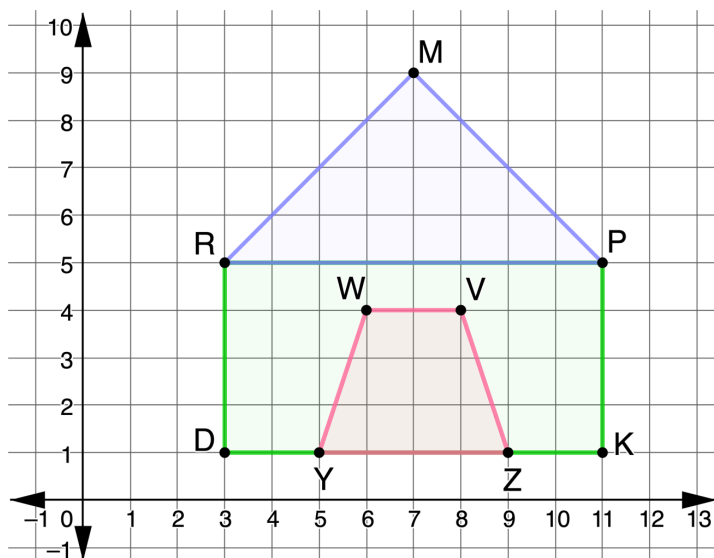
Figure	Perimeter	Area
$\triangle MKR$		
$DMRH$		

- b. What is the perimeter of the composite figure? If necessary, round to the nearest thousandth.
- c. What is the area of the composite figure? If necessary, round to the nearest thousandth.



13.14.2 Collaboration: Real-World Problems

1. Lana is designing the front of a doghouse. The computer software generated coordinate points where each unit equals a foot. The roof has vertices at $M(7,9)$, $P(11,5)$, and $R(3,5)$. The front wall has vertices at $P(11,5)$, $K(11,1)$, $D(3,1)$, and $R(3,5)$. The door entrance has vertices at $Z(9,1)$, $Y(5,1)$, $W(6,4)$, and $V(8,4)$.



- a. What type of triangle is the roof, $\triangle MPR$? Justify your answer.
- b. What is the area of $\triangle MPR$?

- c. Clay tiles cost \$8 per square foot. How much will it cost Lana to put clay tiles on the given face of the doghouse's roof?
- d. What type of quadrilateral is the door, $WVZY$? Justify your answer.
- e. Lana is placing a frame around the door, $WVZY$. How many feet of wood will she need? If necessary, round to the nearest thousandth.
- f. What type of quadrilateral is the front wall, $RPKD$? Justify your answer.

- g. Lana is painting the front wall, $RPKD$, excluding the door, $WVZY$. What is the area of the front wall Lana is painting?

- h. Paint costs \$0.55 per square foot. Excluding the door, how much will it cost to paint the front wall?

2. Justin is creating a layout for his garden in the shape of a quadrilateral, $GWDP$, with vertices $G(1,8)$, $W(11,8)$, $D(11,0)$, and $P(1,0)$.

- a. Classify $GWDP$. Justify your answer.

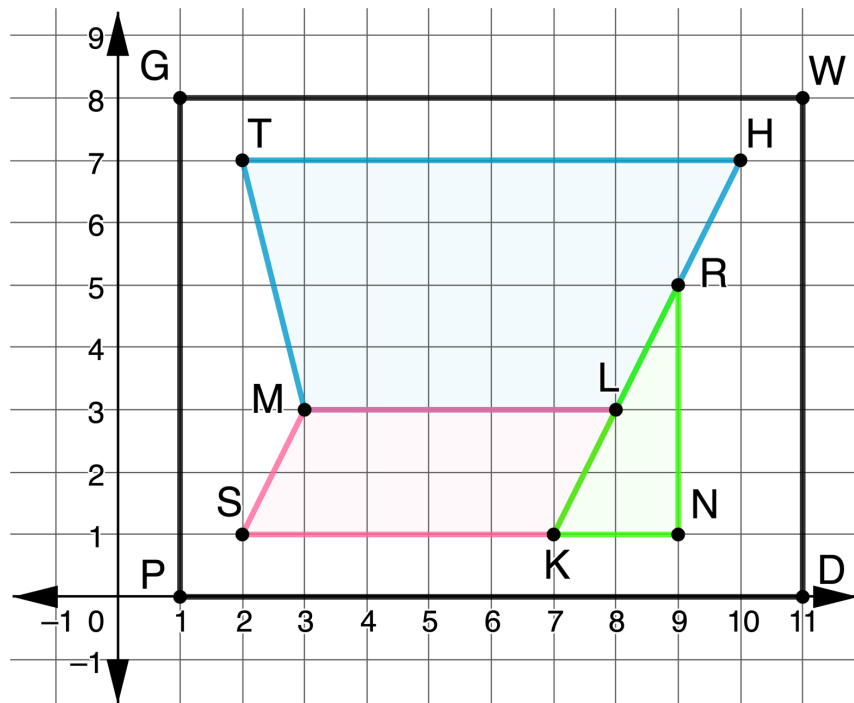
Justin created three sections in the garden: $THLM$, $LKSM$, and $\triangle KNR$.

- b. $THLM$ has vertices at $T(2,7)$, $H(10,7)$, $L(8,3)$, and $M(3,3)$. Classify $THLM$. Justify your answer.

- c. $LKSM$ has vertices at $L(8,3)$, $K(7,1)$, $S(2,1)$, and $M(3,3)$. Classify $LKSM$. Justify your answer.

- d. $\triangle KNR$ has vertices at $K(7,1)$, $N(9,1)$, and $R(9,5)$. Classify $\triangle KNR$. Justify your answer.

The layout of the garden is shown.



e. If one square on the grid represents one square foot, how many feet of fencing will Justin need to go around $GWDP$?

f. Complete the table. If necessary, round to the nearest thousandth.

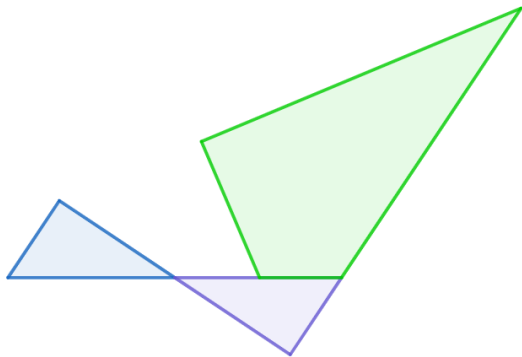
Section	Perimeter	Area
$THLM$		
$LKSM$		
$\triangle KNR$		

g. What is the perimeter of the composite figure made by the three sections of the garden? If necessary, round to the nearest thousandth.

h. What is the area of the three sections of the garden? If necessary, round to the nearest thousandth.

i. If a 40 pound (lb) bag of soil covers 12 square feet, how many bags of soil will Justin need to cover the three sections of the garden?

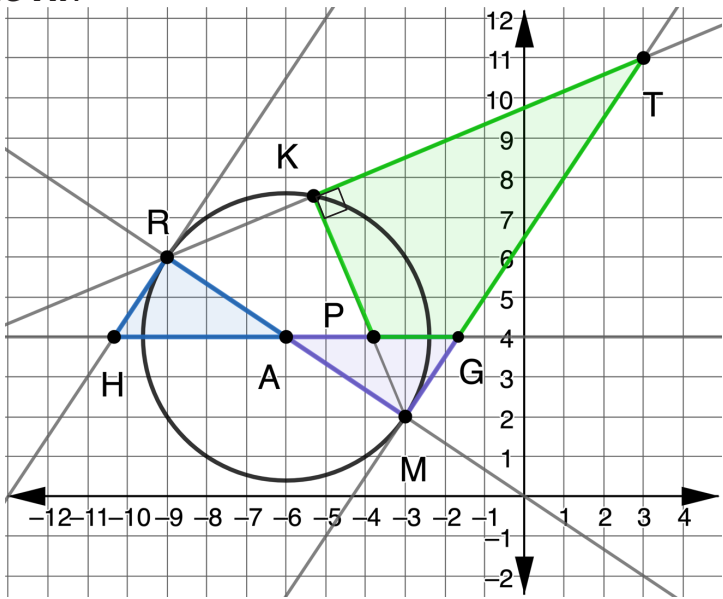
3. Edwin started creating a new logo design for the company office using the images shown.



To determine how to place the logo on the wall in the company's office, he uploaded his design to a computer program and got the following coordinates.

$M(-3,2)$	$R(-9,6)$	$P(-3.8,4)$	$T(3,11)$
$G(-\frac{5}{3},4)$	$H(-\frac{31}{3},4)$	$K(-5.3,7.5)$	$A(-6,4)$

Circle A has points R , M , and K on the circle. Lines TM and HR are tangent to circle A . Lines RM and HG are secant to circle A . Line segment KM is perpendicular to line TK .



- a. Complete the table by finding the area of each triangle.

Triangle	Work	Area
$\triangle AHR$		
$\triangle AMG$		
$\triangle PGM$		
$\triangle TKM$		

- b. Explain how to use $\triangle PGM$ and $\triangle TKM$ to find the area of $TKPG$.
- c. What is the area of $TKPG$? If necessary, round to the nearest thousandth.
- d. What is the total area of Edwin's design? If necessary, round to the nearest thousandth.



13.14.3 Wrap-Up: Reflection

Circle the icon that best describes your understanding of each.



I need help.



I got it.



I could help someone.

Using coordinate geometry to determine if lines are parallel or perpendicular.	  
Using coordinate geometry to partition a line segment or find the weighted average.	  
Using coordinate geometry to classify triangles or quadrilaterals.	  
Using coordinate geometry to justify definitions, properties, and theorems involving circles, triangles or quadrilaterals.	  
Using coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	  
Using coordinate geometry to solve mathematical and real-world involving perimeter or area of polygons.	  

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